

Total No. of Questions : 10]

SEAT No. :

P3103

[Total No. of Pages :6

[5461] - 140

B.E. (Mechanical)

OPERATION RESEARCH

(2012 Course) (Semester - I) (Elective - II) (402045 C) (End Sem.)

Time : 2½ Hours]

[Max. Marks :70

Instructions to the candidates:

- 1) *Answer Q1. or Q2, Q3. or Q4, Q5. or Q6, Q7. or Q8, Q9. or Q10.*
- 2) *Answers in One answer Books.*
- 3) *Figures to the right indicate full marks.*
- 4) *Assume suitable data, if necessary.*

Q1) a) Discuss the various phases in solving an OR problem. **[4]**

- b) Two players P and Q play a game. Each of them has choose one of the three colours, white (W), Black (B) and the red (R) independently of the other. Thereafter the colours are compared. If both P & Q have chosen white (W, W), neither wins anything. If players P select white and player Q selects Black (W, B), player P loses Rs 2 or player Q wins the same amount and so on. The complete payoff matrix is as shown below. Find the optimum strategies for P and Q and the value of the game. **[6]**

		Player Q		
		W	B	R
Player P	W	2	-2	7
	B	2	5	6
	R	3	-3	8

OR

- Q2)** A company has four territories open and four salesmen available for an assignment. The territories are not equally rich in their sales potential. It is estimated that a typical salesman operating in each territory would bring in the following annual sales:

[10]

P.T.O.

Territory	I	II	III	IV
Annual sales (Rs.)	1,26,000	1,05,000	84,000	63,000

The four salesmen also differ in their ability. It is estimated that working under the same conditions, their yearly sales would be proportionality as follows.

Salesmen	A	B	C	D
Propositional	7	5	5	4

If the criterion is maximum, expected total sales, the intuitive answer is to assign the best salesmen to the richest territory, the next best salesmen to the second richest and so on verify this answer by the assignment technique.[10]

- Q3)** a) Discuss the generalised model of linear programming (LPP) in OR. [4]
b) Find out the initial feasible solution by Vogel's Approximation Method (VAM). [6]

Plants or Origins	Destinations				Supply
	1	2	3	4	
1	2	3	11	7	6
2	1	0	6	1	1
3	5	8	15	9	10
Requirement	7	5	3	2	Total=17

OR

- Q4)** Under an employment promotion program, it is proposed to allow sale of newspapers on the buses during off-peak hours. The vendor can purchase the newspapers at a special concessional rate of 25 paise per copy against the selling price of 40 paise. Any unsold copies are, however a dead loss. The vendor has estimated the following probability distribution for the number of copies and demanded as follows: [10]

Weekly Sales	15	16	17	18	19	20
Probability	0.04	0.19	0.33	0.26	0.11	0.07

- a) How many copies should be order so that his expected profit will be maximum?
b) Compute EPPI.
c) The vendor is thinking of spending on a small market survey to obtain additional information regarding the demands level. How much should he be willing to spend on such a survey?

- Q5) a)** Discuss in brief any two Inventory control Techniques. [4]
- b)** An engineering is offered two types of material handling equipments A and B. A is priced at Rs. 5,00,000 including the cost of installation and the cost for the operation and maintenance are estimated as Rs 60,000 for each of the first 5 years and increasing every year by Rs. 20,000 in the sixth and the subsequent year. Equipment B the same rated capacity is priced at Rs. 2,50,000 including the cost of installation and the cost for the operation and maintenance are estimated as Rs 1,20,000 for each of the first 5 years and increasing every year by Rs. 20,000 in the fifth year onwards. The company expected the return of 10% on all its investment. Neglect the scrap value of the equipment at the end of its economic life, determine which equipment the company should buy? [12]

OR

- Q6) a)** A manufacturer has to supply his customers with 600 units of his product per year. Shortage are not allowed and storage amount of 60 paise per unit per year. The set up cost per run is Rs. 80.00. Find out the [8]
- Economic ordering quantity
 - Minimum yearly average cost.
 - Minimum yearly total inventory cost. When the cost per unit item is 2/- per unit.
 - Optimum number of orders per year.
 - The optimum period of supply per optimum order.

The increase in the order cost associated with ordering (1) 20% more than the EOQ.

- b)** The following mortality rates have been observed for a certain type of fuse.

Week	1	2	3	4	5
%failing by the End of Week	5	15	35	75	100

There are 1000 fuses in use and it costs Rs 5 to replace an individual fuse. If all fuses were replaced simultaneously it would cost Rs 1.25 per fuse. It is proposed to replace all fuses at fixed intervals of time, whether or not they have burnt out and to continue replacing burnt out fuses as they fail. At what intervals the group replacement should be made? Also prove that this optimal policy is superior to the straight forward policy of replacing each fuse only when it fails. [8]

- Q7) a)** Six jobs are to be process on three machines. The processing time is as follows, Find the optimal schedule so that the total elapsed time is minimized. [10]

Job	J1	J2	J3	J4	J5	J6
Machine M1 [Turning]	10	3	5	4	2	1
Machine M2 [Threading]	2	4	6	3	1	2
Machine M3[Knurling]	8	6	7	9	7	7

- b) Explain with the help of neat sketch a generalised queuing model. Also explain the Kendall's notation for representing queuing models? [6]

OR

- Q8) a)** A repair shop attended by a single mechanic has an average of four customers an hour who bring small appliances for repair. The mechanic inspects them for defects and takes six minutes an average. Arrivals are Poisson and service rate has the exponential distribution. [8]

You are required to

- Find the proportion of time during which there is no customer in the shop.
 - Find the probability of finding at least one customer in the shop.
 - What is the average number of customers in the system?
 - Find the average time spent by a customer in the shop including service.
- b) Use graphical method to minimize the time needed to process the following jobs on the machine shown i.e. each machine find the job which should be done first. Also calculate the total elapsed time to complete both jobs. [8]

Job (J1)	Sequence	Machine				
		A	B	C	D	E
	Time (hrs)	3	4	2	6	2

Job (J2)	Sequence	Machine				
		B	C	A	D	E
	Time (hrs)	5	4	3	2	6

Q9) a) Differentiate between CPM and PERT [6]

b) A bank has decided to modernize its office. The major elements of the project are as follows. [12]

Activity	Description	Predecessor Activity	Duration (Days)
A	Design New premises	----	14
B	Obtain tenders from contractors	A	4
C	Select the contractor	B	2
D	Arrange details with selected contractor	C	1
E	Decide which equipment is to be used	A	2
F	Arrange storage of equipment	E	3
G	Arrange disposal of other equipment	E	2
H	Order new equipment	E	4
I	Take delivery of new equipment	H,L	3
J	Renovations take place	K	12
K	Remove old equipment for storage or disposal	D,F,G	4
L	Cleaning after the contractor finished	J	2
M	Return old equipment for storage	H,L	2

- Draw an arrow diagram for this project.
- Find out the critical path
- For each non-critical activity find out the total, free and independent float or Slacks.

OR

Q10)a) A project has the following data.

[12]

Task	1-2	1-3	1-4	2-5	2-6	3-6	4-7	5-7	6-7
Optimistic Time	5	18	26	16	15	6	7	7	3
Pessimistic time	10	22	40	20	25	12	12	9	5
Most likely time	8	20	33	18	20	9	10	8	4

Determine the following:

- Construct the network and find out the \Expected task time and their variance.
- The earliest and the latest expected completion times of each event.
- The critical path.
- The probability of an event occurring at the expected completion date if the original scheduled time of completing the project is 41.5 weeks.
- The duration of the project that will have 95% chance of being completed. Use following table.

Probability	0.3085	0.3050	0.3015	0.2981	0.95
Normal distribution	-0.50	-0.51	-0.52	-0.53	1.64
Constant (Z)					

b) Discuss in brief (Any Two).

[6]

- Monte - Carlo Simulation.
- Common Errors/ Flaws in Network.
- Goal Programming.
- Crashing in the network.

